



AHMAD SIDAOU

DATA ENGINEER - DATA INTEGRATION, ANALYTICS & FORECASTING • VIENNA 1220, AUSTRIA • +436642053998

◦ DETAILS ◦

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◦ SKILLS ◦

Python, SQL, Databricks, Airflow,
ETL/ELT, Data Pipelines, Data
Integration, REST APIs

Data Validation, QA/QC
Workflows, Workflow Automation,
Data Cleansing, Data Reliability

Power BI, Spotfire,
Data Modeling, Forecasting,
Operational Analytics, KPI
Monitoring

Docker, Git, C#, PostgreSQL, SQL
Databases

AWS, CI/CD

◦ LANGUAGES ◦

English

French

Arabic

German



PROFESSIONAL SUMMARY

Data Engineer focused on solving operational data challenges through data integration, workflow automation, data quality, and analytics. Over the past 3 years, I have designed and delivered data solutions used to monitor industrial operations, improve data reliability, automate engineering workflows, and support business decision-making. Combining an engineering mindset with hands on experience in Python, SQL, Databricks, Airflow, and Power BI, I build scalable systems that turn complex industrial data into trusted and actionable information.



EMPLOYMENT HISTORY

Data Engineer at Weatherford, Vienna, Austria

Aug 2023 — Mar 2026

- Designed and implemented an hourly data integration bridge between Weatherford's Foresite and PetroVisor platforms using Python, APIs, Airflow, and C#, enabling the transfer of production, SCADA, and well metadata and supporting customer adoption of advanced forecasting capabilities.
- Delivered a production-ready prototype within **2 months**, enabling an Omani oil & gas customer to expand from one Weatherford product subscription to two.
- Developed an automated QA/QC framework validating approximately **20,000 assets** and up to **200,000 production records daily**, replacing manual Excel-based validation and restoring trust in customer forecasting and reporting workflows.
- Built and maintained production analytics workflows processing approximately **700,000 industrial records per day** across more than **20,000 assets** using Databricks, Python, and Airflow.
- Enabled engineers to identify shutdown risks, equipment issues, and production deferment risks through automated data processing, ranking logic, and production forecasting workflows.
- Led the technical delivery of a critical 1-month project for an Omani oil & gas operator, integrating SQL, Oracle, SharePoint, Excel, and PDF-based data sources into a unified operational reporting environment.
- Designed and optimized a Power BI semantic model supporting millions of records, reducing dashboard response and refresh times from several minutes to approximately 30 seconds.
- Delivered seven operational dashboards for a newly commissioned visualization center, contributing to successful PetroVisor adoption and increased customer interest in Weatherford digital solution

Research Assistant at OMV / TU Wien, Vienna, Austria

Mar 2022 — Aug 2023

- Conducted research in enhanced oil recovery and reservoir engineering, analyzing pressure, temperature, geological, and sensor data to evaluate fluid interactions and production optimization strategies.
- Developed Python-based tools and automated workflows for laboratory data collection, processing, and analysis, improving the consistency and accessibility of experimental datasets.
- Processed and analyzed simulation and experimental data using Python, MATLAB,

SQL, Excel, and OpenFOAM, supporting reservoir modeling and engineering studies.

- Worked with geological models, well logs, sensor measurements, and simulation outputs to evaluate reservoir behavior and production performance.
- Introduced modernized approaches for laboratory data management and automated data processing, reducing manual handling and improving data organization for research activities



EDUCATION



Dipl.-Ing. Petroleum Engineering, Montanuniversität Leoben, Leoben, Austria
Feb 2019 — Nov 2022



Bachelor of Engineering - Civil Engineering, Beirut Arab University, Tripoli, Lebanon
Jan 2014 — Nov 2018



PUBLICATIONS



Machine Learning Driven Problem Detection for Production Optimization - Well Cease to Flow Prediction , ADIPEC, UAE
Nov 2024

<https://doi.org/10.2118/222548-MS>



Physics-Informed Reinforcement Learning for Motor Frequency Optimization in Electrical Submersible Pumps: Enhancing AI-Led Decision-Making in Production Optimization , Texas, USA
Aug 2024

<https://doi.org/10.2118/219542-MS>